Electric field

PHYSICS APPLIED
Photocopier

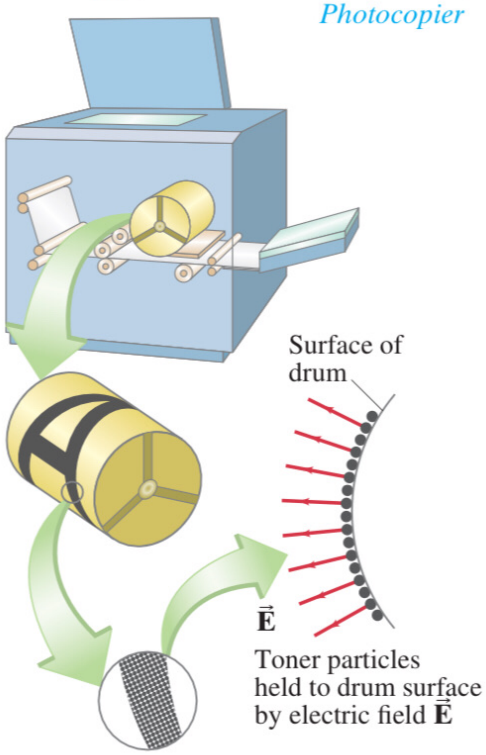
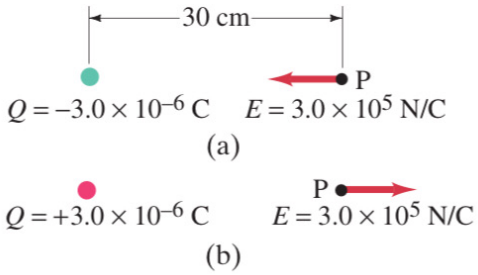


FIGURE 16–25 Example 16–6.

FIGURE 16–26 Example 16–7. Electric field at point P (a) due to a negative charge Q, and (b) due to a positive charge Q, each 30 cm from P.



EXAMPLE 16–6 **Photocopy machine.** A photocopy machine works by arranging positive charges (in the pattern to be copied) on the surface of a drum, then gently sprinkling negatively charged dry toner (ink) particles onto the drum. The toner particles temporarily stick to the pattern on the drum (Fig. 16–25) and are later transferred to paper and "melted" to produce the copy. Suppose each toner particle has a mass of 9.0×10^{-16} kg and carries an average of 20 extra electrons to provide an electric charge. Assuming that the electric force on a toner particle must exceed twice its weight in order to ensure sufficient attraction, compute the required electric field strength near the surface of the drum.

EXAMPLE 16–7 **Electric field of a single point charge.** Calculate the magnitude and direction of the electric field at a point P which is 30 cm to the right of a point charge $Q = -3.0 \times 10^{-6}$ C.

Solutions
16-6

APPROACH The electric force on a toner particle of charge $q = 20e$ is $F = qE$, where E is the needed electric field. This force needs to be at least as great as twice the weight (mg) of the particle.

SOLUTION The minimum value of electric field satisfies the relation

$$qE = 2mg$$

where $q = 20e$. Hence

$$E = \frac{2mg}{q} = \frac{2(9.0 \times 10^{-16}\ \text{kg})(9.8\ \text{m/s}^2)}{20(1.6 \times 10^{-19}\ \text{C})} = 5.5 \times 10^3\ \text{N/C}.$$

16-7

APPROACH The magnitude of the electric field due to a single point charge is given by Eq. 16–4. The direction is found using the sign of the charge Q.

SOLUTION The magnitude of the electric field is:

$$E = k\frac{Q}{r^2} = \frac{(9.0 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)(3.0 \times 10^{-6}\ \text{C})}{(0.30\ \text{m})^2} = 3.0 \times 10^5\ \text{N/C}.$$

The direction of the electric field is *toward* the charge Q, to the left as shown in Fig. 16–26a, since we defined the direction as that of the force on a positive test charge which here would be attractive. If Q had been positive, the electric field would have pointed away, as in Fig. 16–26b.

NOTE There is no electric charge at point P. But there is an electric field there. The only real charge is Q.

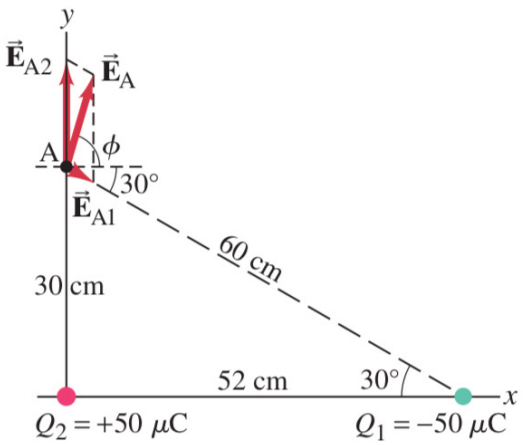


FIGURE 16–28 Calculation of the electric field at point A, Example 16–9.

PROBLEM SOLVING

Ignore signs of charges and determine direction physically, showing directions on diagram

EXAMPLE 16–9 **$\vec{\mathbf{E}}$ above two point charges.** Calculate the total electric field at point A in Fig. 16–28 due to both charges, Q_1 and Q_2.

APPROACH The calculation is much like that of Example 16–4, except now we are dealing with electric fields instead of force. The electric field at point A is the vector sum of the fields $\vec{\mathbf{E}}_{A1}$ due to Q_1, and $\vec{\mathbf{E}}_{A2}$ due to Q_2. We find the magnitude of the field produced by each point charge, then we add their components to find the total field at point A.

SOLUTION The magnitude of the electric field produced at point A by each of the charges Q_1 and Q_2 is given by $E = kQ/r^2$, so

$$E_{A1} = \frac{(9.0 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)(50 \times 10^{-6}\ \text{C})}{(0.60\ \text{m})^2} = 1.25 \times 10^6\ \text{N/C},$$

$$E_{A2} = \frac{(9.0 \times 10^9\ \text{N}\cdot\text{m}^2/\text{C}^2)(50 \times 10^{-6}\ \text{C})}{(0.30\ \text{m})^2} = 5.0 \times 10^6\ \text{N/C}.$$

The direction of E_{A1} points from A toward Q_1 (negative charge), whereas E_{A2} points from A away from Q_2, as shown; so the total electric field at A, $\vec{\mathbf{E}}_A$, has components

$$E_{Ax} = E_{A1}\cos 30° = 1.1 \times 10^6\ \text{N/C},$$
$$E_{Ay} = E_{A2} - E_{A1}\sin 30° = 4.4 \times 10^6\ \text{N/C}.$$

Thus the magnitude of $\vec{\mathbf{E}}_A$ is

$$E_A = \sqrt{(1.1)^2 + (4.4)^2} \times 10^6\ \text{N/C} = 4.5 \times 10^6\ \text{N/C},$$

and its direction is ϕ (Fig. 16–28) given by $\tan\phi = E_{Ay}/E_{Ax} = 4.4/1.1 = 4.0$, so $\phi = 76°$.

Solution
16-9

16–7 and 16–8 Electric Field, Field Lines

19. (I) Determine the magnitude and direction of the electric force on an electron in a uniform electric field of strength 2460 N/C that points due east.

20. (I) A proton is released in a uniform electric field, and it experiences an electric force of 1.86×10^{-14} N toward the south. Find the magnitude and direction of the electric field.

21. (I) Determine the magnitude and direction of the electric field 21.7 cm directly above an isolated 33.0×10^{-6} C charge.

22. (I) A downward electric force of 6.4 N is exerted on a $-7.3\ \mu$C charge. Find the magnitude and direction of the electric field at the position of this charge.

23. (II) Determine the magnitude of the acceleration experienced by an electron in an electric field of 756 N/C. How does the direction of the acceleration depend on the direction of the field at that point?

24. (II) Determine the magnitude and direction of the electric field at a point midway between a $-8.0\ \mu$C and a $+5.8\ \mu$C charge 6.0 cm apart. Assume no other charges are nearby.

25. (II) Draw, approximately, the electric field lines about two point charges, $+Q$ and $-3Q$, which are a distance ℓ apart.

26. (II) What is the electric field strength at a point in space where a proton experiences an acceleration of 2.4 million "g's"?

27. (II) An electron is released from rest in a uniform electric field and accelerates to the north at a rate of 105 m/s^2. Find the magnitude and direction of the electric field.

28. (II) The electric field midway between two equal but opposite point charges is 386 N/C, and the distance between the charges is 16.0 cm. What is the magnitude of the charge on each?

29. (II) Calculate the electric field at one corner of a square 1.22 m on a side if the other three corners are occupied by 3.25×10^{-6} C charges.

30. (II) Calculate the electric field at the center of a square 42.5 cm on a side if one corner is occupied by a $-38.6\ \mu$C charge and the other three are occupied by $-27.0\ \mu$C charges.

31. (II) Determine the direction and magnitude of the electric field at the point P in Fig. 16–56. The charges are separated by a distance $2a$, and point P is a distance x from the midpoint between the two charges. Express your answer in terms of Q, x, a, and k.

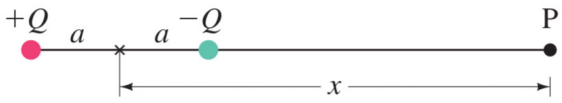


FIGURE 16–56 Problem 31.

Solutions

19. $3.94 \cdot 10^{-16}$ N west
20. $1.16 \cdot 10^{5}$ N/C south
21. $6.30 \cdot 10^{6}$ N/C upward
22. $88 \cdot 10^{5}$ N/C up
23. $1.33 \cdot 10^{14}$ m/s^2
opposite to the field
24. $1.4 \cdot 10^{8}$ N/C
towards the
negative charge
25. 3 times more lines
on the larger charge
26. 0.25 N/C
27. $5.93 \cdot 10^{-10}$ N/C south
28. $1.31 \cdot 10^{-10}$ C
29. $3.36 \cdot 10^{4}$ N/C
30. $1.15 \cdot 10^{6}$ N/C
toward $-38.\mu$C charge
31. $\frac{-4kQx q}{(x^2-a^2)^2}$

32. (II) Two point charges, $Q_1 = -32\,\mu C$ and $Q_2 = +45\,\mu C$, are separated by a distance of 12 cm. The electric field at the point P (see Fig. 16–57) is zero. How far from Q_1 is P?

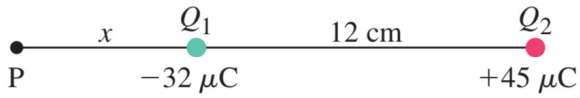


FIGURE 16–57 Problem 32.

33. (II) Determine the electric field $\vec{\mathbf{E}}$ at the origin 0 in Fig. 16–58 due to the two charges at A and B.

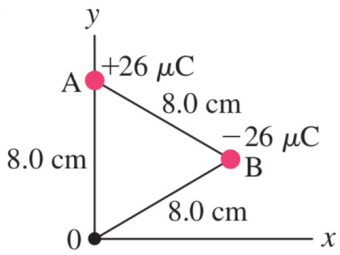


FIGURE 16–58
Problem 33.

34. (II) You are given two unknown point charges, Q_1 and Q_2. At a point on the line joining them, one-third of the way from Q_1 to Q_2, the electric field is zero (Fig. 16–59). What is the ratio Q_1/Q_2?

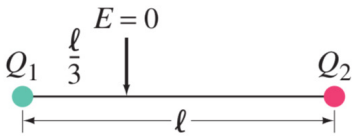


FIGURE 16–59
Problem 34.

35. (III) Use Coulomb's law to determine the magnitude and direction of the electric field at points A and B in Fig. 16–60 due to the two positive charges ($Q = 4.7\,\mu C$) shown. Are your results consistent with Fig. 16–32b?

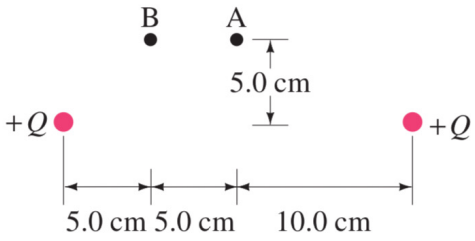


FIGURE 16–60
Problem 35.

36. (III) An electron (mass $m = 9.11 \times 10^{-31}$ kg) is accelerated in the uniform field $\vec{\mathbf{E}}$ ($E = 1.45 \times 10^4$ N/C) between two thin parallel charged plates. The separation of the plates is 1.60 cm. The electron is accelerated from rest near the negative plate and passes through a tiny hole in the positive plate, Fig. 16–61. (*a*) With what speed does it leave the hole? (*b*) Show that the gravitational force can be ignored.

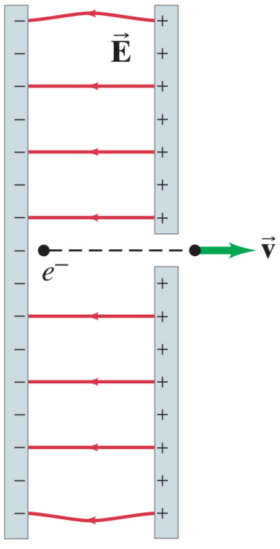


FIGURE 16–61
Problem 36.

Solutions

32. 65 cm

33. $3.7 \cdot 10^7$ N/C

34. $\frac{1}{4}$

35. $E_A = 3.0 \cdot 10^6$ N/C

$\theta_A = 90°$

$E_B = 7.8 \cdot 10^6$ N/C

$\theta_B = 56°$

36. a) $9.03 \cdot 10^6$ m/s

b) $a/g = 2.60 \cdot 10^{14}$